

Henrik Pedersen *AI Software Engineer*

 hnpedersen.tech@gmail.com

 +1 206 758 4994

 North Carolina, United States

 in/hnpedersen

Profile

- Accomplished AI Engineer specialized in designing, developing, and optimizing large language models to solve complex natural language processing tasks, integrating deep learning techniques with scalable systems to build intelligent, language-driven applications.
- As a Backend Engineer, specializes in designing and implementing server-side logic that powers web applications. With expertise in various programming languages and frameworks, ensures seamless integration with databases and external services. A commitment to writing clean, maintainable code optimizes application performance and enhances user experience.

Skills

Programming Language — Python | JavaScript | TypeScript | Golang | SQL

AI Development — LangChain | LangGraph | Chainlit | OpenAI | PyTorch | Tensorflow | TensorRT | LoRA | Pinecone | Milvus | Whisper | ElevenLabs | GeneFace++ | BERT | StackAI | Voiceflow | DialogFlow

Backend Development — Django | Flask | FastAPI | Express.js | Node.js | Nest.js | Gin | Fiber | FastHTTP

Cloud Computing Platforms — AWS | Microsoft Azure | GCP | Lambdalabs | Fly.io

CI/CD & Containerization — Jenkins | GitLab CI/CD | CircleCI | GitHub Actions | Docker | Kubernetes

Testing Frameworks — Pytest | Behave | Comet | Jest | Mocha | Cypress | Ginkgo | Gomega | Scalar | Postman

Database — MySQL | PostgreSQL | Supabase | GraphQL | MongoDB | Neo4j | Redis

Professional Experience

09/2023 – 08/2025

AI Software Engineer

Rogo AI 

Rogo AI offers a software solution designed to facilitate and manage M&A advisory transactions efficiently and securely.

- Designed and developed a fully customized **Chainlit**-based application to serve as the front-end interface for the conversational AI assistant, ensuring an intuitive and responsive user experience tailored to the M&A advisory workflow.
- Engineered the **LangGraph** architecture using modular functional nodes to orchestrate complex, multi-step conversational flows, enabling the chatbot to handle intricate transaction queries with contextual awareness.
- Integrated **Single Sign-On** (SSO) authentication to streamline secure access for existing users, enhancing security while simplifying the user login experience.
- Developed a semantic search engine capable of retrieving and ranking relevant transaction deals using **NLP** techniques, significantly improving the speed and accuracy of information discovery.
- Leveraged **Milvus**, a high-performance vector database, to store embedded representations of transaction data, and integrated it with a **RAG** model to prioritize and surface the most contextually relevant results based on semantic similarity.
- Designed a flexible and scalable session history manager that maintains detailed interaction logs per user, allowing seamless continuity and personalized experiences across sessions.

03/2020 – 08/2023

Generative AI Engineer

Yoodli 

At Yoodli, we're pioneering the future of virtual communication through a photo-realistic, AI-driven assistant, designed to transform video chat experiences. Leveraging a bespoke knowledge base, we ensure that every interaction is as seamless and intuitive as face-to-face conversations.

- Developed an end-to-end pipeline integrating advanced technologies such as speech-to-text, text-to speech, and lip-syncing to bring our vision to life.
- Leveraged an open-source **STT** model **Whisper** to introduce a live captioning feature, enhancing accessibility and understanding in real-time interactions.
- Engineered a system to seamlessly integrate customized knowledge bases with **GPT**, enabling the assistant to deliver personalized and contextually relevant responses.
- Integrated **ElevenLabs** API for **TTS**, and lip-syncing technologies, achieving video generation in under 2 seconds for a virtually lag-free video chat experience.
- Structured the functionality to fine-tune **lip-syncing** model **GeneFace++** specifically for a given avatar.
- Created a robust **FastAPI** server infrastructure to efficiently manage streaming processes for each connection.
- Successfully deployed the entire system on **GCP's Kubernetes**, ensuring scalability and reliability for users worldwide.

02/2019 – 02/2020

AI Backend Engineer

Beacons AI 

Beacons AI is a chat commerce platform for retail and ecommerce to increase sales and conversion. It helps brands to connect all social messengers such as WhatsApp, Line, IG & FB Messenger and provide Sales, Marketing & Customer Service to retail and e-commerce brands.

- Developed an AI-based automatic sales ordering system with high accuracy, aiding customers in product selection and purchasing.
- Incorporated **Twilio** API to enable interactions via mobile applications such as WhatsApp and Line.
- Constructed the entire pipeline using **Voiceflow**.
- Devised an NLP system to automatically categorize emails as spam or legitimate using **BERT**, implementing a web crawler to collect training data for text classification models.

10/2017 – 12/2018

Backend Engineer

Real Time Feedback 

Real Time Feedback is a collaborative diagramming and modeling tool designed to help software engineering and product teams align on technical decisions.

- Implemented services following a service-oriented architecture, encompassing the persistence layer, service layer, and database layer.
- Developed API specifications using OpenAPI Specifications and auto-generated code with swagger commands.
- Involved with bug fixing on functionality, performance and designing issues with JavaScript, jQuery, HTML5, CSS3, React and Angular.
- Enhanced backend system performance by optimizing database queries and reducing latency, employed Redis for efficient caching and refined TypeScript code to improve concurrency.

Education

09/2013 – 08/2017

Bachelor in Computer Science

East Carolina University